

Ex.no:2	EDA-DATA IMPORT AND EXPORT

Aim:

To inspect dataset features, detect null values, and export data into CSV, Excel, and SQLite database formats.

Algorithm:

1. Import pandas and sqlite3 libraries.
2. Read hotel_bookings.csv using `pd.read_csv()`.
3. Inspect data structure using `.head()`, `.tail()`, `.info()`, `.describe()`, and `.isnull().sum()`.
4. Export DataFrame to .csv and .xlsx formats using `.to_csv()` and `.to_excel()`.
5. Connect to hotel_booking.db using sqlite3 and write data into an SQL table via `.to_sql()`.
6. Close the database connection and end execution.

Program:

```
# Import required libraries

import pandas as pd

# Load the Hotel Booking dataset

df = pd.read_csv("hotel_bookings.csv")

# Display the complete dataset

print("===== DATASET =====")

print(df)

# Display first five records

print("\n===== FIRST FIVE ROWS =====")

print(df.head())

# Display last five records

print("\n===== LAST FIVE ROWS =====")

print(df.tail())

# Display dataset information

print("\n===== DATASET INFORMATION =====")

print(df.info())

# Display statistical summary

print("\n===== STATISTICAL SUMMARY =====")

print(df.describe(include='all'))

# Display dataset shape

print("\nDataset Shape:", df.shape)

# Display column names

print("\nColumns:")
```

```
print(df.columns)

# Check missing values
print("\n===== MISSING VALUES =====")
print(df.isnull().sum())

# Export dataset to CSV
df.to_csv("hotel_bookings_export.csv", index=False)
print("CSV file exported successfully!")

# Export dataset to Excel
df.to_excel("hotel_bookings_export.xlsx", index=False)
print("Excel file exported successfully!")

import sqlite3

# Create SQLite database
conn = sqlite3.connect("hotel_booking.db")

# Export DataFrame to SQL table
df.to_sql(
    "hotel_bookings",
    conn,
    if_exists="replace",
    index=False
)
print("Dataset exported successfully to SQLite database!")
conn.close()
```

Output:

----- DATASET -----

	Index	Date	Open	High	Low	Close	Adj Close	Volume
0	NVA	1965-12-31	528.690002	528.690002	528.690002	528.690002	528.690002	0.0
1	NVA	1966-01-03	527.210022	527.210022	527.210022	527.210022	527.210022	0.0
2	NVA	1966-01-04	527.840027	527.840027	527.840027	527.840027	527.840027	0.0
3	NVA	1966-01-05	531.119995	531.119995	531.119995	531.119995	531.119995	0.0
4	NVA	1966-01-06	532.070007	532.070007	532.070007	532.070007	532.070007	0.0
...	...	...	...	...	...	...	...	...
112452	N100	2021-05-27	1241.119995	1251.910034	1241.119995	1247.069946	1247.069946	379696400.0
112453	N100	2021-05-28	1249.469971	1259.209961	1249.030029	1256.599976	1256.599976	160773400.0
112454	N100	2021-05-31	1256.079956	1258.880005	1248.140015	1248.930054	1248.930054	91173700.0
112455	N100	2021-06-01	1254.609985	1265.660034	1254.609985	1258.579956	1258.579956	155179900.0
112456	N100	2021-06-02	1258.489990	1263.709961	1258.239990	1263.619995	1263.619995	148465000.0

112457 rows x 8 columns

----- FIRST FIVE ROWS -----

	Index	Date	Open	High	Low	Close	Adj Close	Volume
0	NVA	1965-12-31	528.690002	528.690002	528.690002	528.690002	528.690002	0.0
1	NVA	1966-01-03	527.210022	527.210022	527.210022	527.210022	527.210022	0.0
2	NVA	1966-01-04	527.840027	527.840027	527.840027	527.840027	527.840027	0.0
3	NVA	1966-01-05	531.119995	531.119995	531.119995	531.119995	531.119995	0.0
4	NVA	1966-01-06	532.070007	532.070007	532.070007	532.070007	532.070007	0.0

----- LAST FIVE ROWS -----

	Index	Date	Open	High	Low	Close	Adj Close	Volume
112452	N100	2021-05-27	1241.119995	1251.910034	1241.119995	1247.069946	1247.069946	379696400.0
112453	N100	2021-05-28	1249.469971	1259.209961	1249.030029	1256.599976	1256.599976	160773400.0
112454	N100	2021-05-31	1256.079956	1258.880005	1248.140015	1248.930054	1248.930054	91173700.0
112455	N100	2021-06-01	1254.609985	1265.660034	1254.609985	1258.579956	1258.579956	155179900.0
112456	N100	2021-06-02	1258.489990	1263.709961	1258.239990	1263.619995	1263.619995	148465000.0

----- DATASET INFORMATION -----

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 112457 entries, 0 to 112456
Data columns (total 8 columns):
#   Column      Non-Null Count  Dtype
---  ---
0   Index       112457 non-null object
1   Date        112457 non-null object
2   Open        110253 non-null float64
3   High        110253 non-null float64
4   Low         110253 non-null float64
5   Close       110253 non-null float64
6   Adj Close   110253 non-null float64
7   Volume      110253 non-null float64
dtypes: float64(6), object(2)
memory usage: 6.94 MB
```

===== STATISTICAL SUMMARY =====

	Index	Date	Open	High	Low	Close	Adj Close	Volume
count	112457	112457	110253.000000	110253.000000	110253.000000	110253.000000	110253.000000	1.102530e+05
unique	14	14731	NaN	NaN	NaN	NaN	NaN	NaN
top	N225	2017-11-03	NaN	NaN	NaN	NaN	NaN	NaN
freq	14500	14	NaN	NaN	NaN	NaN	NaN	NaN
mean	NaN	NaN	7658.515222	7704.372961	7608.000422	7657.545872	7657.351729	1.273975e+09
std	NaN	NaN	9011.478913	9066.638548	8954.506981	9011.510444	9011.608900	4.315783e+09
min	NaN	NaN	54.869999	54.869999	54.869999	54.869999	54.869999	0.000000e+00
25%	NaN	NaN	1855.030029	1864.510010	1843.979980	1855.060059	1854.179565	0.000000e+00
50%	NaN	NaN	5194.089844	5226.279785	5154.049805	5194.750000	5194.750000	4.329000e+05
75%	NaN	NaN	10134.299810	10207.820310	10060.349610	10134.830080	10134.830080	1.734314e+08
max	NaN	NaN	68775.062500	69403.750000	68516.992190	68775.062500	68775.062500	9.440374e+10

Dataset Shape:
(112457, 8)

Columns:
Index(['Index', 'Date', 'Open', 'High', 'Low', 'Close', 'Adj Close', 'Volume'], dtype='object')

===== MISSING VALUES =====

Index 0
Date 0
Open 2204
High 2204
Low 2204
Close 2204
Adj Close 2204
Volume 2204
dtype: int64

CSV file exported successfully!
Excel file exported successfully!
Dataset exported successfully to SQLite database!

```
===== DATASET INFO =====
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 112457 entries, 0 to 112456
Data columns (total 8 columns):
#   Column      Non-Null Count  Dtype
---  -
0   Index       112457 non-null object
1   Date        112457 non-null object
2   Open        110253 non-null float64
3   High        110253 non-null float64
4   Low         110253 non-null float64
5   Close       110253 non-null float64
6   Adj Close   110253 non-null float64
7   Volume      110253 non-null float64
dtypes: float64(6), object(2)
memory usage: 6.9+ MB
```

```
===== MISSING VALUES IN EACH COLUMN =====
Index      0
Date       0
Open       2204
High       2204
Low        2204
Close      2204
Adj Close  2204
Volume     2204
dtype: int64
```

```
===== TOTAL MISSING VALUES =====
13224
```

```
===== ANY MISSING VALUES? =====
True
```

```
===== NON-MISSING VALUES COUNT =====
Index      112457
Date       112457
Open       110253
High       110253
Low        110253
Close      110253
Adj Close  110253
Volume     110253
dtype: int64
```

***** CONSTANT VALUE *****								
	Index	Date	Open	High	Low	Close	Adj Close	Volume
0	NYA	1965-12-31	528.690002	528.690002	528.690002	528.690002	528.690002	0.0
1	NYA	1966-01-03	527.210022	527.210022	527.210022	527.210022	527.210022	0.0
2	NYA	1966-01-04	527.840027	527.840027	527.840027	527.840027	527.840027	0.0
3	NYA	1966-01-05	531.119995	531.119995	531.119995	531.119995	531.119995	0.0
4	NYA	1966-01-06	532.070007	532.070007	532.070007	532.070007	532.070007	0.0

***** MEAN *****								
	Index	Date	Open	High	Low	Close	Adj Close	Volume
0	NYA	1965-12-31	528.690002	528.690002	528.690002	528.690002	528.690002	0.0
1	NYA	1966-01-03	527.210022	527.210022	527.210022	527.210022	527.210022	0.0
2	NYA	1966-01-04	527.840027	527.840027	527.840027	527.840027	527.840027	0.0
3	NYA	1966-01-05	531.119995	531.119995	531.119995	531.119995	531.119995	0.0
4	NYA	1966-01-06	532.070007	532.070007	532.070007	532.070007	532.070007	0.0

***** MEDIAN *****								
	Index	Date	Open	High	Low	Close	Adj Close	Volume
0	NYA	1965-12-31	528.690002	528.690002	528.690002	528.690002	528.690002	0.0
1	NYA	1966-01-03	527.210022	527.210022	527.210022	527.210022	527.210022	0.0
2	NYA	1966-01-04	527.840027	527.840027	527.840027	527.840027	527.840027	0.0
3	NYA	1966-01-05	531.119995	531.119995	531.119995	531.119995	531.119995	0.0
4	NYA	1966-01-06	532.070007	532.070007	532.070007	532.070007	532.070007	0.0

***** MODE *****								
	Index	Date	Open	High	Low	Close	Adj Close	Volume
0	NYA	1965-12-31	528.690002	528.690002	528.690002	528.690002	528.690002	0.0
1	NYA	1966-01-03	527.210022	527.210022	527.210022	527.210022	527.210022	0.0
2	NYA	1966-01-04	527.840027	527.840027	527.840027	527.840027	527.840027	0.0
3	NYA	1966-01-05	531.119995	531.119995	531.119995	531.119995	531.119995	0.0
4	NYA	1966-01-06	532.070007	532.070007	532.070007	532.070007	532.070007	0.0

***** FORWARD FILL *****								
	Index	Date	Open	High	Low	Close	Adj Close	Volume
0	NYA	1965-12-31	528.690002	528.690002	528.690002	528.690002	528.690002	0.0
1	NYA	1966-01-03	527.210022	527.210022	527.210022	527.210022	527.210022	0.0
2	NYA	1966-01-04	527.840027	527.840027	527.840027	527.840027	527.840027	0.0
3	NYA	1966-01-05	531.119995	531.119995	531.119995	531.119995	531.119995	0.0
4	NYA	1966-01-06	532.070007	532.070007	532.070007	532.070007	532.070007	0.0

***** BACKWARD FILL *****								
	Index	Date	Open	High	Low	Close	Adj Close	Volume
0	NYA	1965-12-31	528.690002	528.690002	528.690002	528.690002	528.690002	0.0
1	NYA	1966-01-03	527.210022	527.210022	527.210022	527.210022	527.210022	0.0
2	NYA	1966-01-04	527.840027	527.840027	527.840027	527.840027	527.840027	0.0
3	NYA	1966-01-05	531.119995	531.119995	531.119995	531.119995	531.119995	0.0
4	NYA	1966-01-06	532.070007	532.070007	532.070007	532.070007	532.070007	0.0

Result: